

Proposed Solution

Maximum Marks: 3 Marks

Date	Team ID	Project Name
15 March 2026	Batch 2023-27 ACOE	Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Solution Overview

Rising Waters is a machine learning-powered flood prediction system trained on 5,000 meteorological records. It compares four classification algorithms and deploys the best-performing model (XGBoost at 96.55% accuracy) as a Flask web application on IBM Cloud.

Attribute	Details
Solution Name	Rising Waters – ML-Based Flood Prediction System
Approach	Supervised Machine Learning — Binary Classification (Flood / No Flood)
Dataset	5,000 synthetic meteorological records; 6 input features; binary target
Algorithms Tested	Decision Tree (78.17%), KNN (80.67%), Random Forest (82.00%), XGBoost (96.55%)
Best Model	XGBoost — 96.55% accuracy; saved as floods.save via Joblib
Web Application	Flask app with 6 HTML pages including dedicated Flood and No Flood result pages
Deployment	Docker containerized → IBM Cloud Foundry with manifest.yml
Key Innovation	Separate result pages (result_flood.html / result_noflood.html) with emergency protocols

Model Performance Comparison

S.No	Algorithm	Train Accuracy	Test Accuracy	Deployed
1	Decision Tree	82.40%	78.17%	No
2	K-Nearest Neighbours	85.50%	80.67%	No
3	Random Forest	88.00%	82.00%	No
4	XGBoost	99.20%	96.55%	Yes — floods.save